MOI Ex Prob 4

https://steel-sci.com/assets/section-property-tables.pdf

From Steel Construction Institute online handbook

Example Area Moment of Inertia (MOI) Problem
(from Mechanics of Materials)

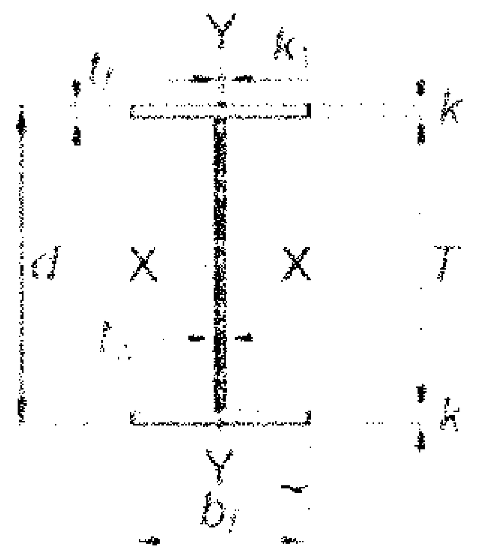
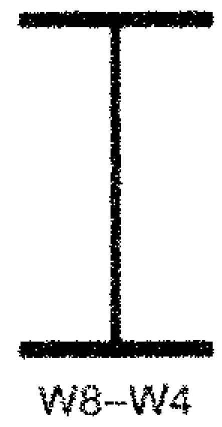


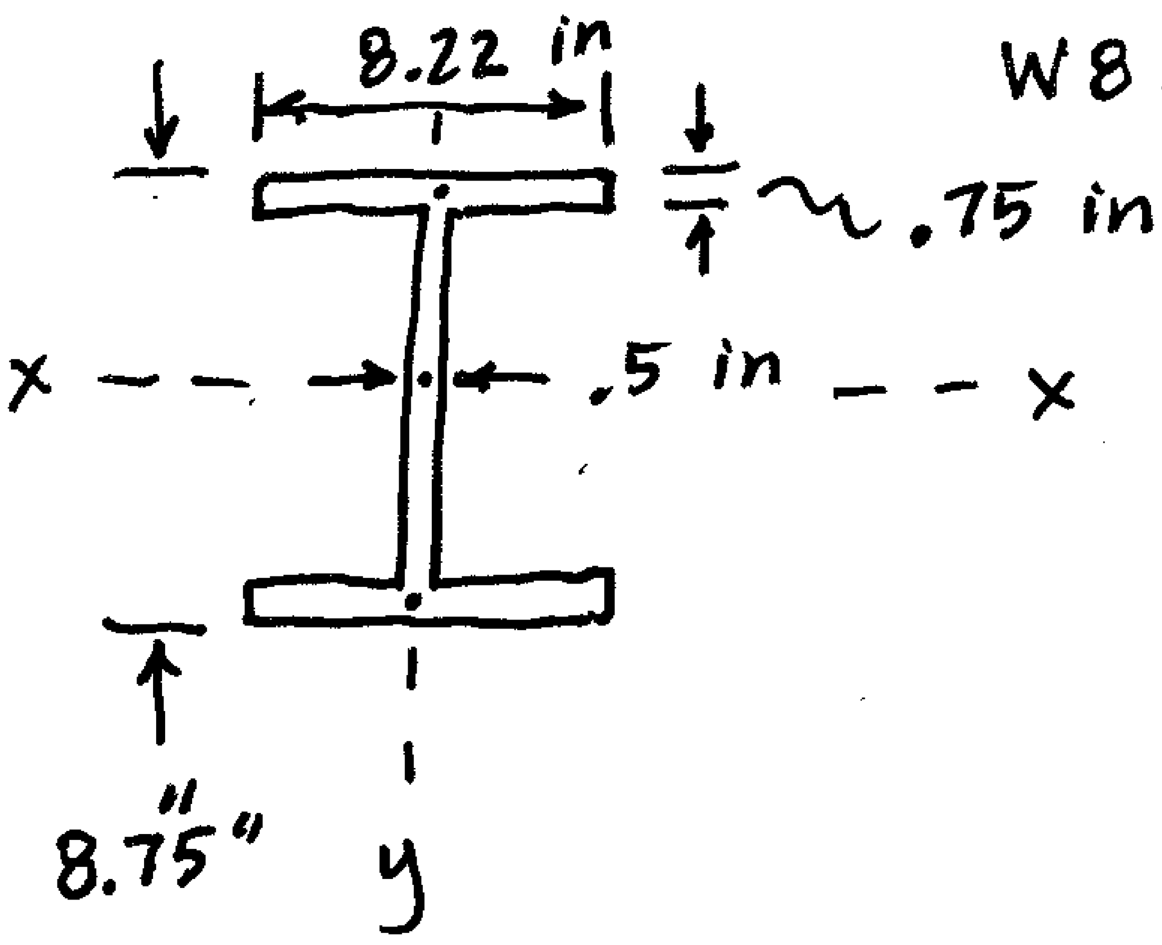
Table 4-1 (continued)
W-Shapes (Welded*)
Dimensions

Shape	Area, A	Nominal Wt.	Depth, d		Web: Thickness, t_w		Web: $\frac{t_w}{2}$	Flange: Width, b_f		Flange: Thickness, t_f	
	in.^{2}	lb/ft	in.		in.		in.	in.		in.	
W8×67	20.5	71.1	9.00	9	0.560	9/16	5/16	8.28	8¼	1.00	1
×58	16.0	55.4	8.75	8¾	0.500	½	¼	8.22	8¼	0.750	¾

Table 4-1 (continued)
W-Shapes (Welded)
Properties



Shape	Compact Section Criteria: $\frac{b_f}{2t_f}$	Compact Section Criteria: $\frac{h}{t_w}$	Axis X-X: I	Axis X-X: S	Axis X-X: r	Axis X-X: Z	Axis Y-Y: I	Axis Y-Y: S	Axis Y-Y: r	Axis Y-Y: Z	r_{ts}	h_o	Torsional Properties: J	Torsional Properties: C_w
			in.^{4}	in.^{3}	in.	in.^{3}	in.^{4}	in.^{3}	in.	in.^{3}	in.	in.	in.^{4}	in.^{6}
W8×67	4.14	12.5	282	62.7	3.71	73.1	94.7	22.9	2.15	34.8	2.46	8.00	5.67	1520
×58	5.48	14.5	214	48.9	3.66	55.9	69.5	16.9	2.09	25.8	2.39	8.00	2.55	1110



W8 × 58 lb/ft (Nominal)

$$I_{xx} = 2\left[\frac{1}{12}(8.22)(.75)^3 + 8.22(.75)(4)^2\right] \text{ Flanges}$$

$$+ \frac{1}{12}(.5)(7.25)^3 \text{ Web}$$

$$\boxed{I_{xx} = 213.7 \text{ in}^4}$$

See 214 in^4 in Table...

$$\frac{8.75}{2} = 4.375$$

$$4.375 - \frac{.75}{2} = 4 \text{ in}$$

$$8.75 - 2(.75) = 7.25$$

$$I_{yy} = 2\left[\frac{1}{12}(.75)(8.22)^3\right] \text{ Flanges}$$

$$+ \frac{1}{12}(7.25)(.5)^3 \text{ Web}$$

$$\boxed{I_{yy} = 69.5 \text{ in}^4}$$

See Table!